Presentation 2

Link to presentation PDF, and code:

Presentation:
https://nuwildcat-my.sharepoint.com/personal/akl0407_ads_northwestern_edu/_layouts/15/stream.aspx?id=%2Fpersonal%2Fakl0407_ads_northwestern_edu%2FDocuments%2FBack%20up%2F2026-27%2FSTAT390%2FStudent%20presentations%2FPresentation%202%2FRecording2_Team3%2Emp4&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2E7928c1ec-0d4c-4926-a1ca-440599cb77a4

PDF:
https://github.com/arvindkrishna87/Team3_DivideNConquer/blob/main/presentation_2/STAT%20390%20-%20Team%203%20Presentation%202.pdf

Code:
https://github.com/arvindkrishna87/Team3_DivideNConquer/tree/main/presentation_2/Code1_AnalyzeResults_5Nov_TaylorMassey

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them.

Ronan (Mathlete): I worked on Mathlete Objective 2. I first looked over the model architecture again and ran some bash commands to find the distribution of the patches of different sizes for the calculations of how many were copies and how many were redundant. After this, I looked back at the two model architecture papers because the use of global pooling sounded very applicable to our model. I then used my calculations along with what I read in these papers to make some recommendations on updating the model architecture. I then asked Claude for any additional updates based on my findings and it recommended the "possible shortcut" on the options slide.

Jingyu & Yifan (Coder): We compared frozen KimiaNet and UNI2-h features for benign versus high-grade classification using the hierarchical attention MIL pipeline. We checked data coverage and grouped splits, extracted UNI2 features, and completed five-split experiments. We reported accuracy, recall for both classes, confusion matrices, and case-level predicted probabilities. We also analyzed loss curves per gradient update and per epoch, identified signs of overfitting, and prepared the comparison and reflection slides.

Taylor (Analyst): I used a script (memorization_results.py) that reloads every saved epoch of the five presentation 1 runs and recomputes each validation case's loss, checking that the recomputed mean matches the validation loss logged during training. I compared each case's errors across presentation 1 validation, presentation 1 test, and

presentation 2 validation. I also compared the presentation 2 split files with those from presentation 1's code and observed that they contain a different number of cases (86 vs. 93); the presentation 1 base code folder I recently copied from the instructional team contains the 86-case files as well. I checked this against the log lines, the project's code walkthrough, and a script. As an outside check, I also compared my recurring error cases with Team 2's Presentation 1 test results and found that their run (different seed and folds) also misclassified 7 of my 9 recurring cases.

How does it help the project?

Ronan: My recommendation helps the project by both pointing out some errors that still exist and if we implement my global pooling recommendation I believe we will save a lot on computational cost.

Jingyu & Yifan: Our comparison evaluates whether changing the frozen feature extractor improves case classification. UNI2 increased pooled test accuracy from 86.02% to 94.62%, with four fewer false positives and four fewer false negatives. Our loss analysis showed improved validation performance in several folds, while identifying folds where overfitting remained. Examining individual predictions also revealed cases where UNI2 introduced new errors despite its overall improvement.

Taylor: It shows that validation loss over only 18-19 cases is dominated by just a few cases, so choosing the best checkpoint or judging memorization from the loss curve alone can sometimes mislead. Accuracy and per-case results are important to report alongside it. My work also identified specific cases that we could have a pathologist review. The case-count check also shows that the presentation 2 split files have fewer benign cases (28 instead of 35), which changes the class balance the weighted loss is meant to correct, so presentation 1 and 2 results should be read with that in mind. A second team's run, with a different seed and folds, also misclassifies 7 of the same 9 cases, which supports that those cases are hard across models rather than specific to my runs.

Issues faced that haven't been resolved:

Ronan: The Optuna script still uses cross entropy loss with mean reduction rather than sum, and I didn't get to manually test some of my recommendations which I definitely want to do over the weekend. I also want to further explore the tuning aspect, since after my findings here I think we can make some significant gains by optimizing our tuning.

Jingyu & Yifan: UNI2's advantage varied across folds, and folds 2 and 5 still showed signs of overfitting. We have not established why some cases remain misclassified or why UNI2 introduces errors in other cases. Our comparison cannot isolate the contributions of pretraining data, architecture, and preprocessing. We also need to

distinguish the supplied 86-case splits from the 93-case experiment reported in our final comparison.

Taylor: The presentation 2 split files contain 86 cases (vs. 93 in my presentation 1 runs), so presentation 1 and 2 results are not directly comparable split by split, and I'm not sure whether this difference is intended. I'm also not entirely sure why cases 26 and 56 are consistently wrong; possible explanations include atypical benign cases, label noise, or overconfidence after memorization, but further testing is necessary. The script's retraining option (run and rescore) has not yet been checked against my original runs; the analysis step reproduces the slides from the saved results.

Proposed next steps:

Ronan: Implement at least some of: global average pooling, both Optuna trained dropout rates, stain dropout, bag-level augmentations, fix the cross entropy loss in optuna_training.py. I plan on seeing which of the above I can manually test over the weekend and seeing their effect on the model. For the next objective, I want to deeply investigate tuning different parameters and how they impact the model, both the hyperparameters in Mathlete Objective 1 and the ones I have suggested here if they are implemented.

Jingyu & Yifan: We plan to archive the exact splits, configurations, checkpoints, and predictions used in the final comparison. We would investigate recurring and newly introduced errors through attention visualizations and data review. Additional experiments with different random seeds could assess whether the improvement persists across partitions. We would also compare feature extraction costs and consider other pathology encoders using the same case grouping and downstream MIL training protocol.

Taylor: Confirm which split files are intended for each presentation. If the 93-case set is intended, regenerate the presentation 2 splits with extra_benign_case_grade_match.csv and add a check that the case count matches the expected total. Run presentation 2's code on presentation 1's 93-case splits as a control. Potentially have a pathologist review a couple of the cases that are consistently wrong. We could also test label smoothing on splits 4 and 5, choose the checkpoint by accuracy or median loss, and retrain split 4 with several seeds to see whether the same cases stay wrong.

References

Whose work did you build up on?

Ronan: Lecture, Code Walkthrough, Presentation 2 base code, Attention-based Deep Multiple Instance Learning (Ilse, et al., 2018), Synergistic H&E and IHC image analysis

by AI predicts cancer biomarkers and survival outcomes in colorectal and breast cancer (Cheng, et al., 2025)

Jingyu & Yifan: Lecture, Code Walkthrough; Lecture, CMIL Classification Problem Statement and Data; Presentation 1 base code; Presentation 2 base code

Taylor: Lecture, Code Walkthrough; Lecture, CMIL Classification Problem Statement and Data; Presentation 1 base code; Presentation 2 base code; Team 2 Presentation 1